

LOAN DEFAULT RISK ANALYSIS

Final Report — Portfolio Insights & Recommendations

1. Executive Summary

This report analyses **9,578 historical loan records** to identify the major factors influencing loan default risk. The dataset includes key borrower characteristics such as **FICO score, annual income, debt-to-income (DTI) ratio, revolving credit utilization, recent credit inquiries, delinquency history, and loan purpose**. Overall, the loan portfolio has a **16.01% default rate**, with borrowers having an average **FICO score of 711** and an average **interest rate of 12.26%**, indicating a portfolio with generally good credit quality.

The analysis reveals that default risk is not evenly distributed across the portfolio but is concentrated within specific borrower segments. Variables such as **loan purpose, FICO score, DTI ratio, revolving credit utilization, and recent credit inquiries** emerge as the strongest predictors of default. These insights provide a foundation for improving **credit underwriting, risk-based pricing, and portfolio monitoring**, enabling lenders to identify high-risk borrowers more effectively while supporting better lending decisions and reducing future credit losses.

10,000	16.01%	711	12.26%
Total Loans	Overall Default Rate	Avg. FICO Score	Avg. Interest Rate

2. Dashboard Overview

To present the analysis in a clear and interactive format, the data was consolidated into a single dashboard that summarizes the portfolio from both a high-level and detailed perspective. The dashboard combines key performance indicators (KPIs), interactive filters, and multiple visualizations covering **loan purpose, FICO score bands, debt-to-income (DTI) bands, revolving credit utilization, recent credit inquiries, and credit policy status**. This layout enables users to quickly understand the overall portfolio while exploring the factors that contribute to loan default.

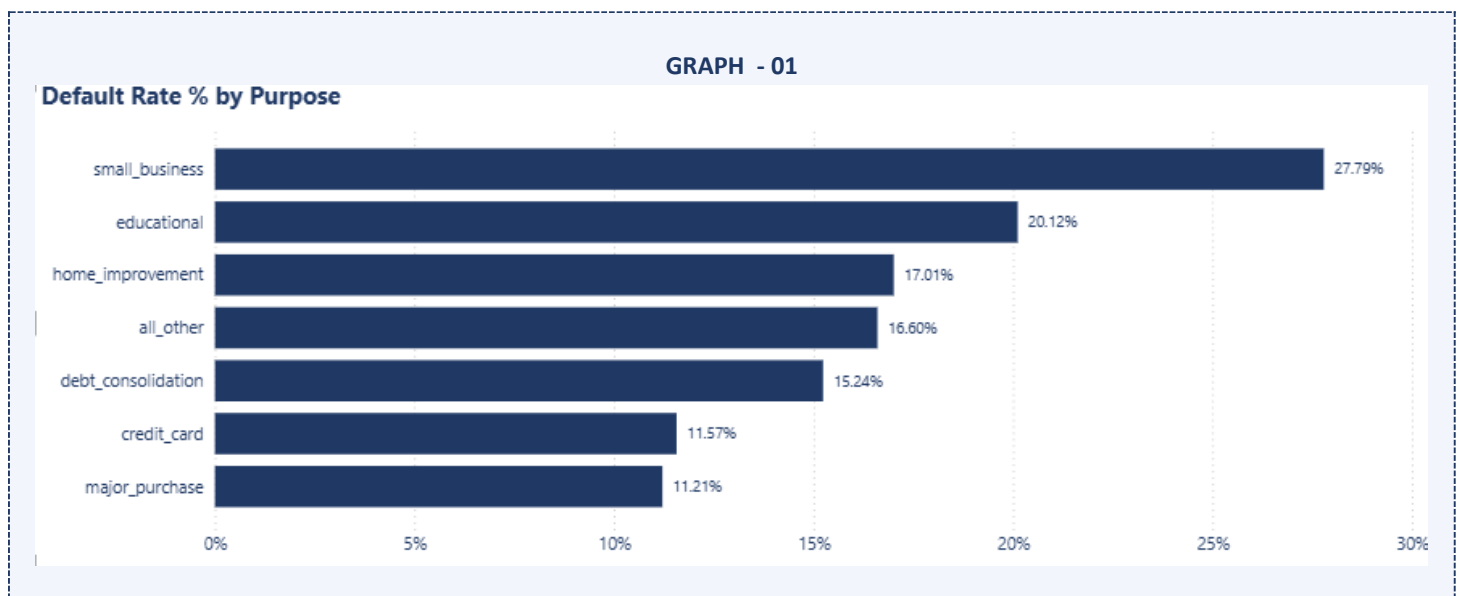
The interactive filters allow users to focus on specific borrower segments and compare their performance across different risk dimensions. By moving from the overall portfolio view to detailed visualizations, lenders can identify high-risk groups, evaluate default patterns, and make more informed decisions regarding underwriting, risk-based pricing, and portfolio monitoring. The dashboard therefore serves as an effective decision-support tool for understanding borrower behaviour and improving credit risk management.

3. Default Rate by Loan Purpose

Loan purpose is one of the most significant factors influencing default risk in this portfolio. Default rates vary considerably across loan categories, ranging from **11.21% for major purchase loans** to **27.79% for small business loans**, making small business lending the highest-risk segment. Educational loans also record a relatively high default rate of **20.12%**, suggesting that borrowers in these categories are more likely to experience repayment difficulties than the average borrower.

On the other hand, **major purchase** and **credit card** loans have the lowest default rates, remaining well below the overall portfolio average of **16.01%**. Loan categories such as **debt consolidation, home improvement, and other purposes** fall closer to the portfolio average, representing moderate levels of risk. These findings demonstrate that loan purpose is an effective indicator of borrower risk and should be considered when designing **underwriting policies, risk-based pricing strategies, and portfolio monitoring practices** to improve lending decisions and reduce future credit losses.

Purpose	Default Rate
Small Business	27.79%
Educational	20.12%
Home Improvement	17.01%
All Other	16.60%
Debt Consolidation	15.24%
Credit Card	11.57%
Major Purchase	11.21%



4. Default Rate by FICO, DTI & Revolving Utilization

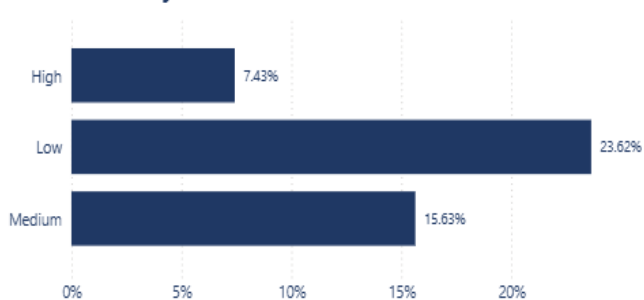
FICO score is the strongest predictor of loan default in the portfolio. Borrowers in the **Low FICO** band have a default rate of **23.62%**, compared with only **7.43%** for the **High FICO** band, while the **Medium FICO** group records **15.63%**. This highlights the strong relationship between credit score and repayment risk.

A similar pattern is observed for **DTI ratio** and **revolving credit utilization**. Borrowers with higher debt burdens and higher credit utilization consistently show higher default rates than those with lower levels. Together, these factors provide a reliable basis for improving underwriting, risk-based pricing, and credit risk management.

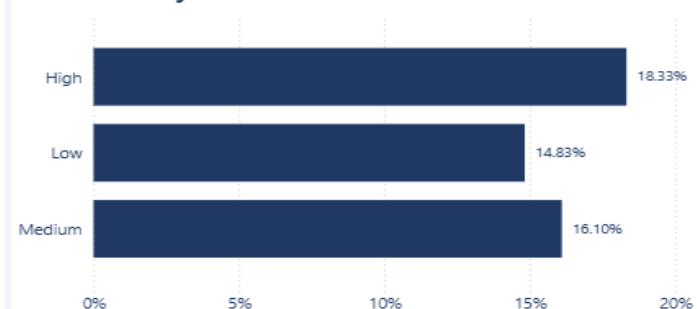
Segment	Band	Default Rate
FICO Band	Low	23.62%
FICO Band	Medium	15.63%
FICO Band	High	7.43%
DTI Band	High	18.33%
DTI Band	Medium	16.10%
DTI Band	Low	14.83%
Revolving Utilization	High	18.97%
Revolving Utilization	Medium	16.33%
Revolving Utilization	Low	12.43%

GRAPH – 02 , 03, 04

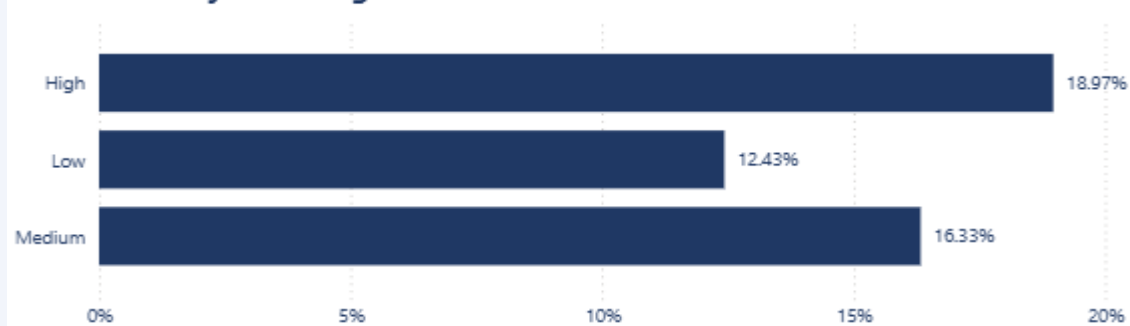
Default Rate % by FICO Band



Default Rate by DTI Band



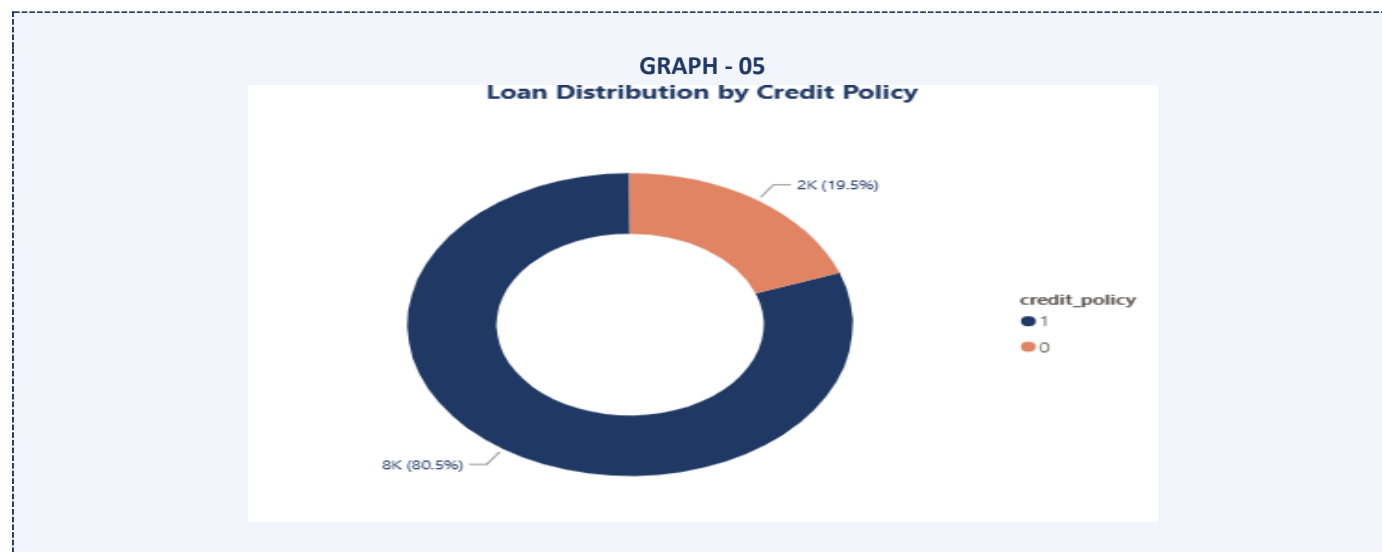
Default Rate by Revolving Utilization



5. Credit Policy Mix

The portfolio consists of **80.5% policy-compliant loans** and **19.5% policy exception loans**. Policy exception loans represent borrowers who were approved despite not fully meeting the lender's standard credit criteria.

Comparing the performance of these two groups helps evaluate the effectiveness of the credit policy. This analysis supports better underwriting decisions, risk-based pricing, and overall credit risk management.



6. Recent Credit Inquiries

The number of **recent credit inquiries** has a clear impact on loan default risk. Borrowers with fewer inquiries have lower default rates, while default rates increase as inquiry counts rise.

Although borrowers with very high inquiry counts represent a small sample, the overall trend shows that frequent credit inquiries are a useful early indicator of higher credit risk and should be considered during loan approval.



7. Key Insights

The analysis identifies a small number of borrower characteristics that consistently explain loan default risk across the portfolio. These findings provide a clear basis for improving lending decisions and strengthening credit risk management.

- **Loan purpose, FICO score, and recent credit inquiries** are the strongest predictors of loan default.
- **Small business (27.79%)** and **educational (20.12%)** loans have the highest default rates and require stricter underwriting.
- **Low-FICO borrowers** default at **23.62%**, compared with **7.43%** for high-FICO borrowers, making credit score the most reliable risk indicator.
- Borrowers with **high DTI (18.33%)** and **high revolving credit utilization (18.97%)** also exhibit above-average default risk.
- Borrowers with **10 or more recent credit inquiries** show significantly higher default rates and should undergo additional credit assessment.
- The **19.5% policy-exception** loan segment should be monitored separately to evaluate the effectiveness of underwriting policies over time.

8. Recommendations

Based on the findings of this analysis, the following recommendations can help reduce loan defaults while maintaining healthy portfolio growth. These actions focus on strengthening underwriting practices, improving risk assessment, and applying more effective pricing strategies.

- Apply **risk-based pricing** for **small business** and **educational loans**, as these categories have the highest observed default rates.
- Use **minimum FICO score** and **maximum DTI thresholds** for automatic approvals, while routing borderline applications for manual review.
- Flag borrowers with **more than 5–6 recent credit inquiries** for additional verification before approving a loan.
- Monitor **policy-exception loans** separately and review their performance regularly to assess the effectiveness of current underwriting standards.
- Include **revolving credit utilization** along with **FICO score** and **DTI ratio** when evaluating borrower risk.
- Develop a **composite credit risk score** by combining **FICO, DTI, revolving utilization, and recent credit inquiries** to improve default prediction and support more accurate lending decisions.